

## Úloha 3.2

$$(a) \ u_1 = \frac{1}{\int_1^5 1 dx} = \frac{1}{4}$$

$$u'_2 = x - \int_1^5 \frac{x}{4} dx = x - 3 \rightarrow u_2 = \frac{x-3}{\int_1^5 (x-3)^2 dx} = \frac{3(x-3)}{16}$$

$$u'_3 = x^2 - \int_1^5 \frac{x^2}{4} dx - \int_1^5 \frac{3(x-3)x^2}{16} dx = x^2 - \frac{49}{3} \rightarrow u_2 = \frac{x^2 - \frac{49}{3}}{\int_1^5 (x^2 - \frac{49}{3})^2 dx} = \frac{45(x^2 - \frac{49}{3})}{15376}$$

$$(b) \ sin(x) = \int_1^5 sin(x) \frac{45(x^2 - \frac{49}{3})}{15376} dx \frac{45(x^2 - \frac{49}{3})}{15376} + \int_1^5 sin(x) \frac{3(x-3)}{16} dx \frac{3(x-3)}{16} + \int_1^5 sin(x) \frac{1}{4} dx \frac{1}{4}$$

$$a = \int_1^5 sin(x) \frac{45(x^2 - \frac{49}{3})}{15376} dx \approx -0.065933$$

$$b = \int_1^5 sin(x) \frac{3(x-3)}{16} dx \approx -0.64656$$

$$c = \int_1^5 sin(x) \frac{1}{4} dx \approx 0.06416$$

$$(c) \ \|sin(x) + 0.065933 \frac{45(x^2 - \frac{49}{3})}{15376} + 0.64656 \frac{3(x-3)}{16} - 0.06416 \frac{1}{4}\|^2 = \int_1^5 (sin(x) + 0.065933 \frac{45(x^2 - \frac{49}{3})}{15376} + 0.64656 \frac{3(x-3)}{16} - 0.06416 \frac{1}{4})^2 dx \approx 1.59139$$